

Network Monitoring and Programming Using Python



Python scripts and APIs can be tailor made into effective network monitoring and forensics tools. Their versatility makes them ideal in assorted applications including cyber security, data mining, Internet of Things, cloud simulation, grid implementation, etc.

and highly effective programming languages, which include Python, Java, PERL, PHP and many others.

Python is one of the widely used languages for writing the special scripts for packet capturing, classification and machine learning.

It should be mentioned that a great deal of network monitoring and logging software has been developed in Python. Shinken and Zenoss are common tools used for monitoring the hosts, network data collection, alerts and messaging, and include lots of active and passive monitoring methods. Currently, Shinken, based on Python, is the open source framework used for monitoring. This software can perform a large set of operations related to digital forensics and logging.

Python scripts and libraries for network forensics

Here's a list of tools built with Python for network monitoring, logging, high security credential management and performance evaluation.

Eddie

The features of this tool include:

- System monitoring checks
- File system checking
- HTTP checks
- POP3 tests
- SNMP queries
- RADIUS authentication tests
- Monitoring of processes

Network monitoring and digital forensics are some of the prominent areas in the domain of cyber security. There are a number of software products and tools available in the technology market to guard network infrastructure and confidential data against cyber threats and attacks.

For a long time, the monitoring of servers and forensic analysis of network infrastructure has been done using packet capturing (PCAP) tools and intrusion detection systems (IDS). These activities are performed using PCAP and IDS tools available in the market, which include open source software as well as commercial products.

Despite the number of tools available for packet capturing and monitoring, professional programmers prefer to use their own software developed by coding and scripting. Self-developed and programmed code offers a lot of flexibility in customising the tool. Many organisations concerned about security, confidentiality and integrity, choose not to use any third party software. Rather, they develop their own tools using efficient